# Algebraic Structures.

## Binary operations:

combine two elements of the set to produce another element of the same set.

## Binary operators

### Properties of Binary operations

i) Closure property
ii) Associative property
iii) Identity property
iv) Inverse property
v) Commutative property
vi) Idempotent.

+, −, ×, ÷ operators.

Closure = algebraic structure
closure + Associative = Semigroups
closure + Associative + Identity = Monoids
closure + Associative + Identity + Inverse = Groups
closure + Associative + Identity + Inverse + commutative = A Abelian group.

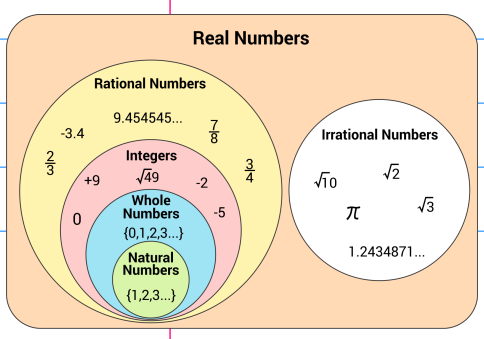
## Algebraic Structure

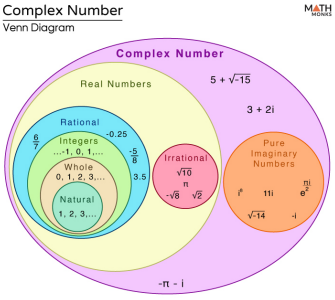
↳ closure property.

Let A be non empty set. An operation * is said to be closure on A if $\forall a, b \in A \quad a * b \in A$

e.g; N, + (closure → yes) $7+1=8$ Z + $-2+5=3$ (yes) R + yes
N, − non closed Z − yes R − yes
N, × yes Z × yes R × yes
N, ÷ No Z ÷ No R ÷ No

↓
in case of division, in real if a number is divided by zero the answer is undefined. So, in case of real number answer should in b/w −∞ and +∞ inclusive.



Semigroups $(A, *)$ be an algebraic structure where $*$ is a binary operation on A. $(A, *)$ is called semigroup if the following conditions are satisfied.

- closure property
- Associative property $(a*b)*c = a*(b*c)$

e.g; $N, +$ yes $\quad Z+$ yes $\quad R+$ yes

$3+(2+5) = 9 \quad (3+2)+5 = 10$

$Z-$ No $\quad R-$ NO

$N, \times$ yes $\quad Z\times$ yes $\quad R\times$ yes

$(1\times 2)\times 3 = 6$

$1\times(2\times 3) = 6$

$(-2-3)-5 = -10$

$-2-(3-5) = 0$

$(-2\times 3)\times 7 = -24$

$-2(3\times 7) = -24$

Monoids $(A, *)$ be an algebraic structure, where $*$ is a binary operation on A. $(A, *)$ is called Monoid. If the following conditions are satisfied.

1. closure property
2. Associative property $a*e = e*a = a$.
3. Identity

e.g; $N, +$ NO $\quad Z+$ yes $\quad R, +$ yes

$3+e=3 \quad -2+e=0 \quad \frac{5}{9}+e=\frac{5}{9}$

$e=0 \quad e=2 \quad e=0$

$N, \times$ yes $\quad Z, \times$ yes $\quad R, \times$ yes

$3\times e = 3 \quad -2\times e = -2 \quad \frac{5}{9}\times e = \frac{5}{9}$

$e=1 \quad e=1 \quad e=1$

**Group** $(A, *)$ be an algebraic structure where $*$ is a binary operation on $A$. $(A, *)$ is called Group if the following conditions are satisfied

1. Closure property
2. Associative property
3. Identity
4. Inverse

e.g.;

$N_i \times q \times a^{-1} = e$ → NO

$q \times a^{-1} = 1$

$a^{-1} = \frac{1}{q}$

$2+ -3 + a^{-1} = e$ ← yes

$-3 + a^{-1} = 0$

$a^{-1} = 3$

$2\times$ ← NO

$-3 \times a^{-1} = e$

$-3 \times a^{-1} = 1$

$a^{-1} = \frac{1}{-3}$

$R, + \rightarrow$ yes

$R, \times \rightarrow$ NO

Abelian group ← yes

$2+$

$-3+2 = -1$

$2+(-3) = -1$

$a * b = b * a$

$R+ \rightarrow$ yes

$\frac{5}{2} + \frac{3}{2} = \frac{8}{2} = 4$

$\frac{3}{2} + \frac{5}{2} = 4$